

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
H2 Biology Preliminary Examinations 2018

H2 BIOLOGY
 Practical

9744/04
03 Aug 2018
2 hours and 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number, class, shift and laboratory on this Question Paper.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/21
2	/20
3	/14
Total	55

This question paper consists of 19 printed pages.

Answer **all** questions.

- 1** You are required to investigate the effect of enzyme concentration on the rate of photosynthesis using plant extract after they have been incorporated into alginate beads.

The rate of photosynthesis can be measured by the changes in colour of an indicator solution which is indicative of the amount of carbon dioxide in the solution. When carbon dioxide dissolves in water, carbonic acid is formed, thus reducing the pH and changing the colour of the indicator solution. The indicator also changes colour when carbon dioxide is removed from water.

At atmospheric concentration of carbon dioxide, this indicator is deep red in colour.

You are provided with:

- Sodium alginate, **A**
- Indicator solution, in a container labelled **B**
- Calcium chloride solution, in a container labelled **C**
- Section of a fresh leaf

Before starting the experiment, read through Steps 1 to 12.

Proceed as follows.

- 1** Place 2g of a fresh leaf on a white tile and carefully remove any large veins. Chop the rest of the leaf into very fine pieces and place the pieces into a boiling tube.
- 2** Using a dropper, add 7cm³ of water to the boiling tube. Carefully grind the chopped leaf in the boiling tube using a glass rod for about 5 minutes to obtain a dark green leaf extract. Pour the leaf extract into a beaker.
- 3** Using a 5cm³ syringe, transfer 3cm³ of the plant extract into a vial, taking care not to introduce air bubbles into the container. Next, transfer 3cm³ of the sodium alginate, **A**, into the vial with the plant extract, again taking care not to introduce air bubbles into the container.
- 4** Gently mix the plant extract and alginate by stirring the mixture with a glass rod. Do not mix vigorously as this may introduce air bubbles into the mixture.

- 5 Attach an empty 10cm³ syringe without its plunger vertically in the clamp stand, as shown in Fig. 1.1.

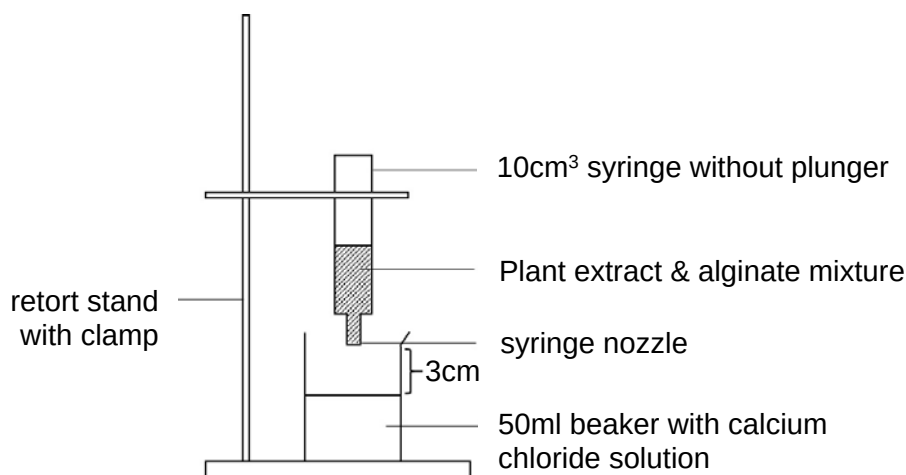


Fig. 1.1

- 6 Pour approximately 30cm³ of calcium chloride, **C** into a 50ml beaker. Place the beaker directly underneath the syringe as shown in Fig. 1.1, ensuring a distance of around 3cm between the syringe nozzle and the surface of the calcium chloride solution.
- 7 Pour the plant extract and alginate mixture into the syringe. Let all the mixture drip into the beaker of calcium chloride. The mixture will form a bead upon contact with calcium chloride. Leave the beads in the beaker of calcium chloride for 5 minutes to let them solidify.
- 8 Using a spoon, transfer the beads into an empty beaker, and rinse them using distilled water. Next, pour away the distilled water and add fresh distilled water. Leave the beads in fresh distilled water.
- 9 Using a syringe, transfer 5cm³ of indicator solution, **B** into a boiling tube. Using a pair of forceps, add 2 alginate beads containing plant extract of similar size and shape into the boiling tube. Quickly cap the boiling tube with a stopper.
- 10 Place the boiling tube into a rack and position the boiling tube 10cm from a lit lamp for 10 minutes.
- (a) (i) State and explain one modification you would make to steps 1 to 10 to increase the degree of confidence in your results.

[2]

- (ii) Suggest a suitable control for this experiment.

[1]

- (iii)** Sketch a fully-labelled graph to show the expected relationship between enzyme concentration and the rate of photosynthesis.

[2]

11 Using a glass rod, stir the mixture in the boiling tube well to ensure that the colour of the solution is homogeneous. Next, use the colour chart to identify the pH of the mixture.

12 Repeat steps **9** and **10** with 6, 10, 14 and 18 alginate beads containing plant extract.

- (iv)** Record your results in a suitable format in the space below.

[3]

- (v) Explain whether your experimental results support the relationship predicted in (iii).

[2]

- (vi) Explain the effect of the number of alginate beads containing plant extract on the rate of photosynthesis.

[3]

- (vii) State and explain one assumption which you made in order for a colour change to be observed at step 11.

[2]

A student carried out the same experiment as the one above except that he used a spectrophotometer to measure the colour change instead of a colour chart.

He recorded his results in Table 1.

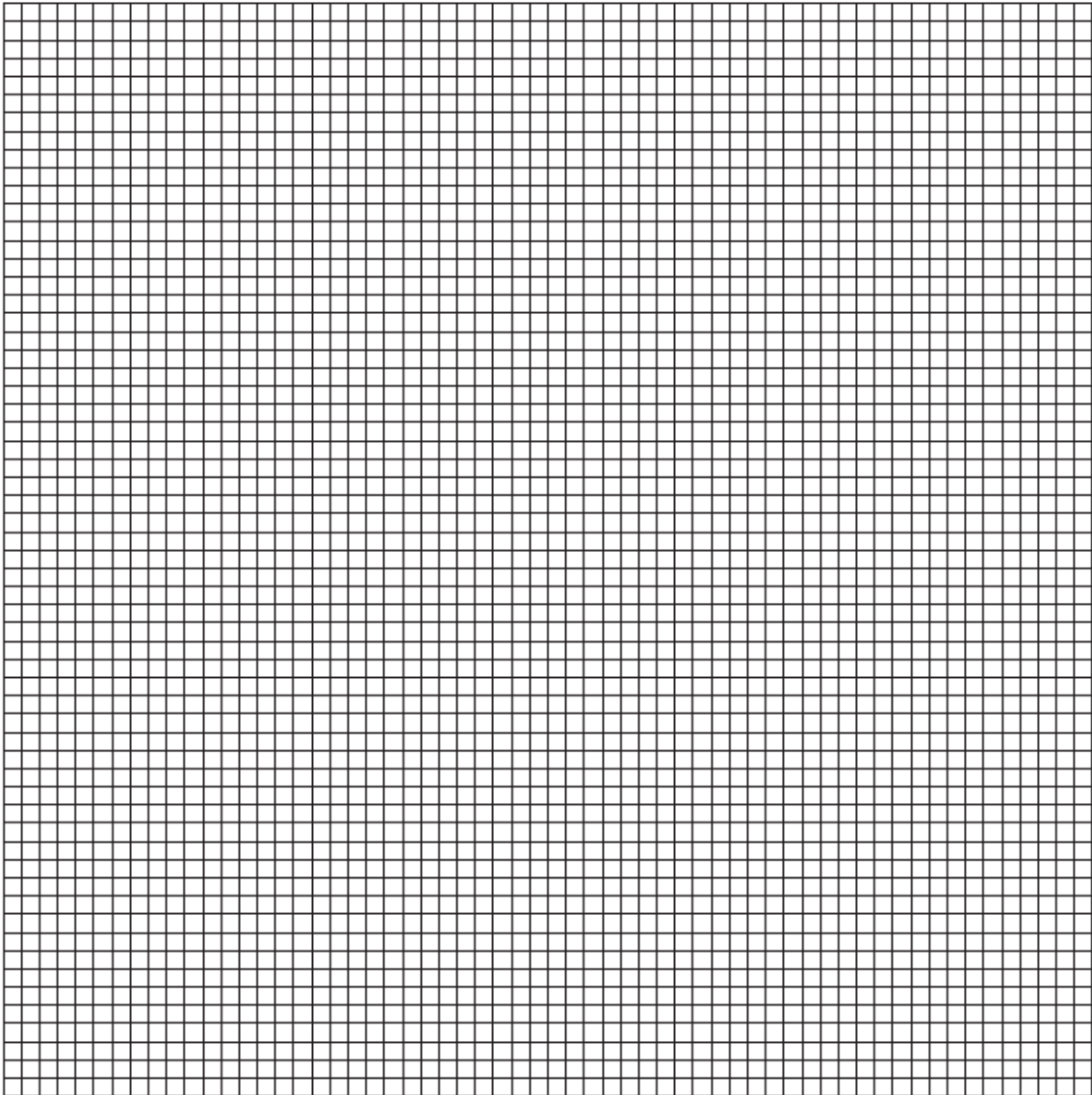
Table 1

Number of alginate beads	Absorbance / Au		
	Replicate 1	Replicate 2	Average
2	0.3655	0.3745	
6	0.6470	0.6500	
10	0.7020	0.7100	
14	0.7415	0.7485	
18	0.7620	0.7690	

(b) (i) Complete Table 1 to show the values of the average results.

[1]

- (ii) Draw a graph of the student's results in Table 1 on the following grid to show the effect of enzyme concentration on absorbance.



[3]

- (iii) Explain why the experiment is more accurate when a spectrophotometer, instead of a colour chart, is used to measure the colour change.

[1]

- (iv) A spectrophotometer needs to be calibrated before it is used to measure the absorbance of the solutions from the above experiment. Describe what you would use to calibrate the spectrophotometer so as to ensure a valid set of readings.

[1]

[Total: 21 marks]

2 Section A

In this section, you will require access to a microscope and a slide **K6**.

K6 is a stained, transverse section of a leaf blade from a plant.

Examine slide **K6** under high-power objectives (x40) of the microscope.

- (a) Use the space provided below to draw a plan diagram of part of the leaf section on slide **K6** as shown in the boxed region in Fig. 2.1. Label your plan diagram appropriately.

You should locate a suitable region of the specimen to be able to observe the distinct categories of different types of cells in the cross section.

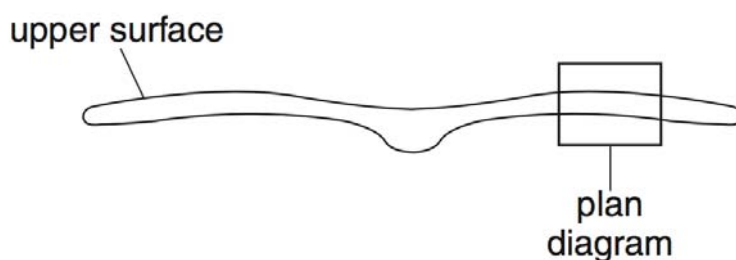


Fig. 2.1

[4]

Fig. 2.2 is a photomicrograph of a stained transverse section of a leaf blade from a **different** species of plant, **K7**.

You are not expected to be familiar with this specimen.

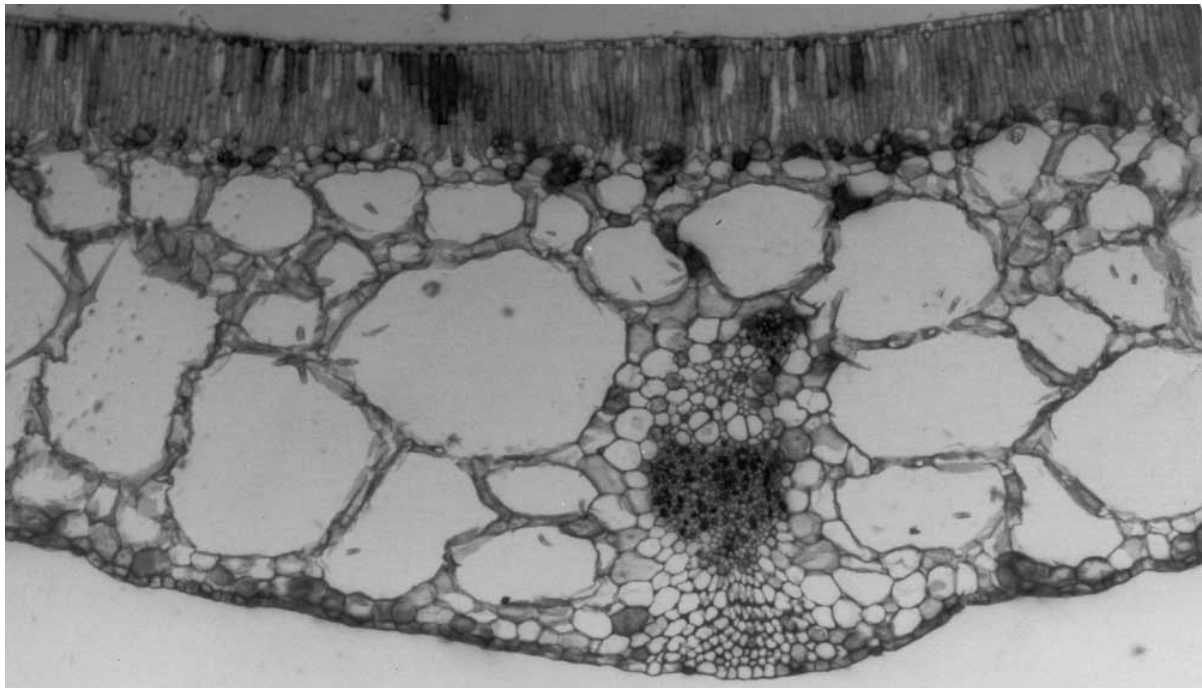


Fig. 2.2

- (b)** Use a suitable table to record two observable differences between the specimen on slide **K6** and **K7**.

- (c) Based on your answer from (b), suggest the types of environment the two species of plant can be found in.

[1]

- (d) Plants with the same tissue arrangement in their leaves seen in **K7** are abundant in the type of environment they are found in.

Suggest how the abundance of these plants supports Darwin's theory of evolution.

[3]

Section B

In this section, you will require access to a microscope.

You will investigate the density of stomata on the lower surface of a leaf.

You are provided with a leaf labelled **L**.

Proceed as follows:

- 1** Cover a small area (about 1cm²) of the lower epidermis of **L** with a **thin** layer of clear nail varnish. Avoid areas with large veins. Repeat this three more times on different areas of the lower epidermis to ensure that you have spare material if necessary.
 - 2** Put **L** to one side for **at least** 15 minutes to allow the nail varnish to dry.
 - 3** After allowing the nail varnish to dry, carefully use the blade of a fine scalpel to lift one edge of the layer of nail varnish. You may then use forceps to gently peel the layer of nail varnish from the leaf. Ensure that the area of the layer obtained is sufficient for the field of view under the microscope. Transfer the layer to a microscope slide. Gently place a cover slip on the thin layer. No water is required to mount the specimen.
- (a) (i)** Examine the slide under high-power objective lens (x40) and locate a stoma.

Make a detailed, labelled drawing in the space below of the stomatal pore, its guard cells and four adjacent epidermal cells.

[4]

- (ii) Calculate the **mean** number of stomata per field of view for **L** under the low-power objective lens (x10).

Show your working.

Mean number of stomata per field of view for **L** =

[1]

- (iii) Using your results from (a)(ii) and given the formula

$$\text{Density of stomata (no. of stomata per mm}^2\text{)} = \frac{\text{No. of stomata}}{\text{Area of field of view,}}$$

calculate the mean density of stomata per mm² for **L**.

Show your working.

Mean density of stomata of **L** =

mm⁻²

[1]

Sun plants are plants that grow in full sunlight and are adapted to relatively high light intensities. Shade plants are not regularly exposed to full sunlight and are adapted to relatively low light intensities.

Leaves of these plants are adapted to differing light intensities. This may be shown by the density of stomata (number of stomata in a square millimeter) on the leaves.

The mean stomata density on leaves from a sun plant and a shade plant was recorded and a statistical t-test was conducted to determine whether there was a significant difference between these means.

A summary of the results is shown in Table 2.

Table 2

Mean stomatal density / mm ⁻²		Significance of difference	Total sample size
Shade leaves	Sun leaves		
290	335	$p < 0.05$	10

(b) (i) Comment on what these results show and suggest an explanation for any pattern.

[3]

(ii) Describe how you can increase the statistical confidence of your conclusions.

[1]

[Total: 20 marks]

3 Planning Question

Bacteria have the ability to acquire new genes via horizontal gene transfer. One of the processes of horizontal gene transfer is the process of transformation. The process of transformation has been utilised by scientists to introduce foreign DNA into bacterial cells for gene cloning purposes.

Escherichia coli cells are treated with calcium chloride solution to take up plasmid DNA containing the ampicillin gene using the heat-shock method. This involves subjecting the bacterial cells to rapid changes in temperature for a brief period of time to create transient pores in the cell membrane. During the heat-shock method, *E. coli* cells are transferred to a water bath at a specific temperature for 50 seconds before immediately returning them to ice for 2 additional minutes.

You are required to plan, but not carry out, an investigation to compare the effect of the heat-shock temperature between 30°C and 50°C on the efficiency of bacterial transformation.

Transformation efficiency can be calculated using the following equation:

$$\text{Transformation efficiency} = \frac{\text{Total number of colonies on agar plate}}{\text{Total number of colonies of viable bacteria}}$$

*cfu: colony forming unit

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- 320µl *Escherichia coli* cells in microfuge tube, on ice
- 10µl solution of plasmid DNA containing ampicillin resistance gene
- LB nutrient broth
- Agar plates with and without ampicillin
- Transformation solution (calcium chloride solution)
- Microcentrifuge tubes (1.5 ml)
- Micropipettes and tips
- Thermostatically controlled water bath
- Sterile distilled water
- Stopwatch
- Sterile spreaders
- Parafilm (to seal Petri dish)
- Incubator oven
- 70% ethanol

You may select from the following apparatus:

- normal laboratory glassware e.g. test tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- syringes
- autoclave (a pressurised oven for heat sterilising apparatus and materials),
- Bunsen burner

Your plan should:

- have a clear and helpful structure such that the method you use is repeatable by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 14 marks]

[illegible]

For
Examiner's
Use

[Turn over]

[Turn over]